Yuan Yuan

Los Angeles, CA | milleryuan0107@gmail.com | 267 593 5897 | Website | Linkedin | Github | Google Scholar

Education

University of California, Los Angeles Sep 2026 – May 2031 (Expected)
- Ph.D. in Computer Science

University of Pennsylvania Sep 2023 – May 2026
- M.S. Data Science & Computer and Information Science GPA: 3.88/4.0

Vassar College Sep 2017 – May 2021
- B.A., Major in Political Science, Correlate in Math GPA: 3.93/4.0

Research Interests

Large Language Model, LLM Agent, Multiagent System, Multimodal LLM, LLM Post-Training & Reinforcement Learning, Synthetic Data

Selected Research

Multi-Agent Inference: Coordination, Exploration, and Reliable Evaluation
Work in Progress
- Built a **multi-agent inference framework** integrating native Codex and Claude Code harnesses with local Qwen models served through vLLM and hosted DeepSeek models. Implemented four-agent shared-memory, independent, and coordinator-led workflows with dynamic delegation, persistent sessions, and controlled candidate access.
- Designed a **12-condition KernelBench LeNet-5 study** across two models, two harnesses, and three coordination architectures, with fixed one-hour runs, token accounting, and reproducible evaluation. Audited **3,764 tool calls across six orchestrator runs** and replayed all winning submissions, uncovering evaluator-source leakage and incomplete correctness checks that confounded apparent performance gains.

Real2Sim Bench: Evaluating Metric 3D Reconstruction from Real-World Images and Video
Work in Progress
- Developing a benchmark of multimodal coding agents' ability to reconstruct metric-scale 3D objects and scenes from real images and video. Following a **hard-to-generate, easy-to-evaluate** design, it uses hidden physical measurements to assess dimensions and spatial relationships rather than visual plausibility alone.
- Built a pilot pipeline with measurement-free input videos, evaluator-only reference videos, agent-generated Blender reconstructions, and **absolute-scale dimension scoring** without post-hoc resizing. Curated real-world captures and reference datasets with provenance and leakage controls; extending evaluation to component geometry, clearances, and deformable objects.

Publications

ImplicitPersona: Continuous Learning of Implicit User Personas via Reinforcement Fine-tuning for LLM Personalization
KDD 2027
*Bowen Jiang, **Yuan Yuan**, Zhuoqun Hao, Zhangchen Xu, Anvesh Rao Vijjini, Ziyi Liu, Radha Poovendran, Dan Roth, Camillo J. Taylor, Sihao Chen.*

Know Me, Respond to Me: Benchmarking LLMs for Dynamic User Profiling and Personalized Responses at Scale
COLM 2025 & Long abstract at the MASC-SLL 2025 [Oral]
Bowen Jiang, Zhuoqun Hao*, Young-Min Cho, Bryan Li, **Yuan Yuan**, Sihao Chen, Lyle Ungar, Camillo J. Taylor*, Dan Roth*.*

A Concise Agent is Less Expert: Revealing Side Effects of Using Style Features on Conversational Agents
Under Review of ARR January
Young-Min Cho, **Yuan Yuan***, Sharath Chandra Guntuku, Lyle Ungar.*

AgentWorld: A Multi-Agent Research Platform for Collaborative AI Systems
COLM 2026
Raphael Zhu, Jeffrey Cho, Yusen Zhang, Wenliang Zheng, ***Yuan Yuan****, Jin Mo Yang, Chi Wang, Rui Zhang.*

One Model, All Roles: Multi-Turn, Multi-Agent Self-Play Reinforcement Learning for Conversational Social Intelligence
arXiv preprint arXiv:2602.03109, 2026
Bowen Jiang, Taiwei Shi, Ryo Kamoi, ***Yuan Yuan****, Camillo J. Taylor, Longqi Yang, Pei Zhou, Sihao Chen.*

TurnaboutLLM: A Deductive Reasoning Benchmark from Detective Games
EMNLP 2025 Main
Yuan Yuan*, Muyu He*, Muhammad Adil Shahid, Jiani Huang, Ziyang Li, Li Zhang.*

ControlText: Unlocking Controllable Fonts in Multilingual Text Rendering without Font Annotations
EMNLP 2025 Findings
Bowen Jiang,* ***Yuan Yuan****, Xinyi Bai, Zhuoqun Hao, Alyson Yin, Yaojie Hu, Wenyu Liao, Lyle Ungar, Camillo J. Taylor.*

Multilingual Retrieval Augmented Generation for Culturally-Sensitive Tasks: A Benchmark for Cross-lingual Robustness
ACL 2025 Main
Bryan Li, Fiona Luo, Samar Haider*, Adwait Agashe*, Tammy Li†, Runqi Liu†, Muqing Miao†, Shriya Ramakrishnan†,* ***Yuan Yuan****†, Chris Callison-Burch.*

Towards Rationality in Language and Multimodal Agents: A Survey
NAACL 2025 Main
Bowen Jiang, Yangxinyu Xie, Xiaomeng Wang, ***Yuan Yuan****, Zhuoqun Hao, Xinyi Bai, Weijie J. Su, Camillo J. Taylor, Tanwi Mallick.*

Professional Experience

AI Engineer, Microsoft – Remote Sep 2025 – Dec 2025

- Experimented with Supervised Finetuning (SFT) and Reinforcement Finetuning (RFT) to internalize LLM-as-a-Judge responses into model parameters, using **verl**.
- Fine-tuned Large Language Models with **RFT** on personalization benchmarks on Azure Cloud.
- Developed and debugged **inference pipelines** for open- and closed-source LLMs, monitoring performance on personalization benchmarks.
- Maintained an internal multi-party **LLM chat platform** built with **FastAPI** and **HTML** for real-time collaboration.
- **Deployed and integrated new models** into the platform, ensuring seamless user interaction and stability.
- Extended model capabilities by implementing **GPT-style function calling** for enhanced reasoning and tool use.

Software Engineer Intern, Meta – Seattle, US May 2025 – Aug 2025

- Built and shipped features for the **Brand Safety Advertiser Experience Team**, collaborating with cross-functional product designers, PMs, and engineers.
- **Initiated and scoped the first AI Agent project** for the team; designed and implemented an internal **Oncall Domain Agent** using Hack and Context Engineering, improving oncall efficiency and developer productivity.
- Enhanced **millions** of advertisers' experience with Facebook Ads by **improving UIs in JavaScript** and **streamlining workflows**, significantly reducing friction in ad interactions.
- Developed an **ads matcher in Hack/PHP** to automatically detect and filter ads based on advertiser requirements, increasing accuracy of ad delivery and covering over **20 million** ads on platforms.

Open Source Maintainer, AsyncAPI at Google Summer of Code – Remote May 2024 – Sep 2024

- Led the development of a **Slack bot** using **Django**, utilizing **AWS** for infrastructure management and **MongoDB** with vector search capabilities; the bot integrated message history, GitHub repository data, and website content to enhance search and retrieval functionalities.
- Led the **deprecation of the AsyncAPI Generator CLI**, transitioning to a more robust and integrated AsyncAPI CLI environment, enhancing platform functionality and streamlined development processes.
- Initiated and directed the **integration of templates into the generator repository**, improving developers' experience and code reusability.
- Used **Git** for version control and code reviews; resolved critical issues in **React** engine templates, improving

functionality and reliability.

Open-Source Contributions

AsyncAPI (Google Summer of Code Maintainer): Led the migration of the Generator CLI and integrated modular templates to enhance developer experience.

- **PRs:** #1251, #1234, #1213 **Issue:** #1249

vLLM (Contributor): Participated in architectural discussions regarding distributed inference reliability and multi-node synchronization.

- **Issue:** #33034 (Proposed a benchmark-driven auto-tuning solution to replace hard-coded Mamba kernel configurations with a dynamic JSON-based loader)

Honors

PHI BETA KAPPA Membership
The Julia Flitner Lamb Prize at Vassar College
Ann Cornelisen Fund for Post-Graduate Fellowships

TECHNICAL SKILLS & FRAMEWORKS

Java, Python, R, C++, Spring Boot, Django, AWS EC2, AWS S3, AWS Kinesis, AWS RDS, AWS Aurora, AWS Elasticache, Redis, Apache Kafka, MySQL, Postgres, DynamoDB, MongoDB, Docker, Docker-compose, Kubernetes, Jenkins, ArgoCD, gRPC, HTTP, WebSocket